

The Use of the Integral Method to Compare the Efficiency of Neural Networks to Improve Image Resolution Depending on Size and Format Using the Example of Road Photos

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Abstract— Roads occupy one of the most important places in the life of a modern person. The quality of the road surface is its key characteristic. To assess the quality of roads, there are many different systems that mainly solve the problem using image analysis. For better work with photos, they are preprocessed. Preprocessing, for example, includes changing the image resolution, which can be implemented using machine learning methods, including the use of neural networks. In this paper, a comparative analysis of the neural networks efficiencies to improve image resolution, depending on size and format, using the integral method is carried out. The neural networks selected are EDSR, ESPCN, ESRGAN, FSRCNN, LapSRN. The research material consists of road surface images in 5 sizes (100x100, 141x141, 173x173, 200x200, 224x224) and 3 formats (bmp, jpg, png). All neural networks have a linear $O(n)$ dependence of the average time to obtain an improved image version on the photo size for each format. EDSR, ESPCN, FSRCNN, LapSRN also have a linear $O(n)$ dependence for the average allocated memory on the size for all formats. ESRGAN has a constant $O(1)$ dependence on the allocated memory for jpg and png formats, and almost constant dependence for bmp. ESPCN has the best time efficiency, and ESRGAN has the best space efficiency. ESRGAN and ESPCN also have the best integral efficiency values.

Keywords—neural network, image resolution, efficiency, integral method, size, format, road photo

I. INTRODUCTION

Roads occupy one of the most important places in the life of a modern person. The quality of the road surface is its key characteristic. To assess the quality of roads, there are many different systems that mainly solve the problem using image analysis. For better work with photos, they are preprocessed. Preprocessing, for example, includes changing the image resolution, which can be implemented using both traditional image processing algorithms [1-3] and machine learning methods [4-7], which include the use of neural networks.

In this paper, an analysis of existing neural networks is carried out. Since already trained neural networks have been selected, it makes no sense to consider such characteristics as precision, recall and accuracy for their analysis. Only computational characteristics are considered – the average image processing time (time indicator) and the average value of the allocated memory (space indicator). The most efficient neural network is one that successfully combines both indicators. The method combining time and space indicators is called the integral.

The **purpose** of the work is to analyze the efficiency of neural networks in improving image resolution using the integral method.

The **research material** is road surface photos in various sizes and formats.

The **scientific novelty** lies in the integral approach for evaluating the efficiency of neural networks.

The research objectives:

1. To provide brief information about the neural networks selected for analysis to improve image resolutions.
2. To describe the methodology of conducting an experiment to obtain performance indicators of the analyzed neural networks based on the integral method.
3. To analyze the results obtained and draw the appropriate conclusions from the study.

II. NEURAL NETWORKS FOR IMAGE SUPER RESOLUTION

For the experiment to evaluate the efficiency of neural networks using the integral method, the following are selected: EDSR, ESPCN, ESRGAN, FSRCNN, LapSRN. Each neural network increases the resolution in height and width by 4 times (the number of pixels changes by 16 times). These neural networks are used for analysis because they are easy to

implement and provide the opportunity to conduct experiments in the Visual Studio Code development environment. Brief information about each neural network is provided below.

EDSR – enhanced deep super-resolution network [8]. The model [9] is used for the experiment.

ESPCN – efficient sub-pixel convolutional neural network [10]. A trained neural network is used for the analysis [11].

ESRGAN – enhanced super-resolution generative adversarial network [12]. For the experiment, a trained model from the Kaggle website is used [13].

FSRCNN – fast super-resolution convolutional neural network [14]. The model [15] is used for the analysis.

LapSRN – Laplacian pyramid super-resolution network [16]. A trained neural network is used for the experiment [17].

III. METHODOLOGY

For the experiment, 10 road surface photos for 5 different sizes (100x100, 141x141, 173x173, 200x200 and 224x224) in bmp, jpg and png formats are used. The image sizes are selected in such a way that the number of pixels in the photo varies evenly. The listed sizes in pixels: 10000 (100x100), 20000 (≈141x141), 30000 (≈173x173), 40000 (200x200), 50000 (≈224x224). The change step is 10000 pixels. The experimental computer characteristics: operating system – Windows 10 Pro, system type – x64, RAM – 8 GB, RAM type – DDR4, RAM channel – single, hard drive type – basic. The latest version of Visual Studio Code has been selected as the development environment [18], the application type is Jupyter-notebook [19], and Python is the programming language (version 3.11.3) [20]. The library “time” [21] is used to record time, and “tracemalloc” [22] is used to measure memory.

The experiment to determine the efficiency of neural networks to improve image resolution has the following stages:

1. The images described above are used as research material for each of the analyzed neural networks (EDSR, ESPCN, ESRGAN, FSRCNN, LapSRN).
2. For 5 selected neural networks, 10 different images are improved for each of the 3 formats and 5 resolutions. 10 images for each experiment are stored in a separate computer folder (75 different folders in total: 5 (neural networks) x 3 (formats) x 5 (resolutions)). The improved versions of the images are placed in a new folder located in the same location as the experimental one.
3. For each experiment (improving the resolution of 10 different photos), the total execution time (in seconds) and the total allocated memory (in megabytes) are recorded. To find their average values, each of the listed indicators is divided by the number of images in the folder (10 in total). The formulas are presented below.

$$t_{avr} = t_{total}/N \quad (1)$$

where t_{avr} is average time to obtain an improved version of the image, t_{total} is total time spent for one experiment, N is number of experimental photos (equal to 10).

$$s_{avr} = s_{total}/N \quad (2)$$

where s_{avr} is the average memory used to obtain an image improved version, s_{total} is the total memory used during one experiment.

IV. RESULTS AND DISCUSSION

An example of photos and their improved versions is shown in Fig. 1.



Fig. 1. An example of photos and their super resolution versions.

The average time and memory spent on obtaining an improved image version in various formats and resolutions for each neural network are shown in Tables 1 and 2 respectively.

TABLE I. AVERAGE TIME (IN SECONDS) TO OBTAIN AN IMPROVED IMAGE VERSION IN VARIOUS FORMATS AND RESOLUTIONS FOR SELECTED NEURAL NETWORKS

BMP					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	3.710	8.297	14.298	22.159	29.638
ESPCN	0.016	0.028	0.039	0.058	0.088
ESRGAN	3.488	12.784	22.301	30.986	40.035
FSRCNN	0.018	0.036	0.066	0.098	0.114
LapSRN	0.396	0.853	1.299	1.799	2.269
JPG					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	4.110	9.183	14.931	22.406	28.793
ESPCN	0.024	0.039	0.045	0.064	0.080
ESRGAN	4.132	13.145	22.277	30.361	39.747
FSRCNN	0.020	0.048	0.086	0.108	0.112
LapSRN	0.411	0.880	1.324	1.790	2.276
PNG					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	4.121	9.495	15.095	22.550	29.056
ESPCN	0.157	0.190	0.265	0.354	0.463
ESRGAN	3.794	12.553	21.893	31.208	39.017
FSRCNN	0.079	0.229	0.279	0.383	0.473
LapSRN	0.514	1.024	1.557	2.030	2.562

TABLE II. AVERAGE ALLOCATED MEMORY (IN MEGABYTES) WHEN OBTAINING AN IMPROVED IMAGE VERSION IN VARIOUS FORMATS AND RESOLUTIONS FOR SELECTED NEURAL NETWORKS

BMP					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	500.593	991.848	1492.472	1993.978	2500.836
ESPCN	499.245	991.361	1491.775	1993.346	2500.165
ESRGAN	129.642	76.143	77.328	76.906	77.132
FSRCNN	499.112	991.274	1491.664	1993.241	2500.036
LapSRN	499.149	991.298	1491.779	1993.313	2500.285
JPG					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	499.423	991.854	1492.318	1993.908	2500.788
ESPCN	499.236	991.234	1491.694	1993.227	2500.039
ESRGAN	5.165	5.507	5.544	5.694	5.468
FSRCNN	499.058	991.248	1491.633	1993.199	2500.088
LapSRN	499.094	991.255	1491.687	1993.319	2500.132
PNG					
Neural network	Pixels				
	10000	20000	30000	40000	50000
EDSR	499.414	991.804	1492.346	1993.924	2500.748
ESPCN	499.188	991.183	1491.632	1993.203	2500.076
ESRGAN	5.198	5.639	5.455	5.479	5.684
FSRCNN	499.094	991.177	1491.614	1993.274	2500.005
LapSRN	499.159	991.254	1491.680	1993.318	2500.185

To make the results presented in Tables I and II more visual, corresponding graphics of the dependence of average time and allocated memory on resolution and format have been plotted. The graphics are demonstrated in Fig. 2 and Fig. 3.

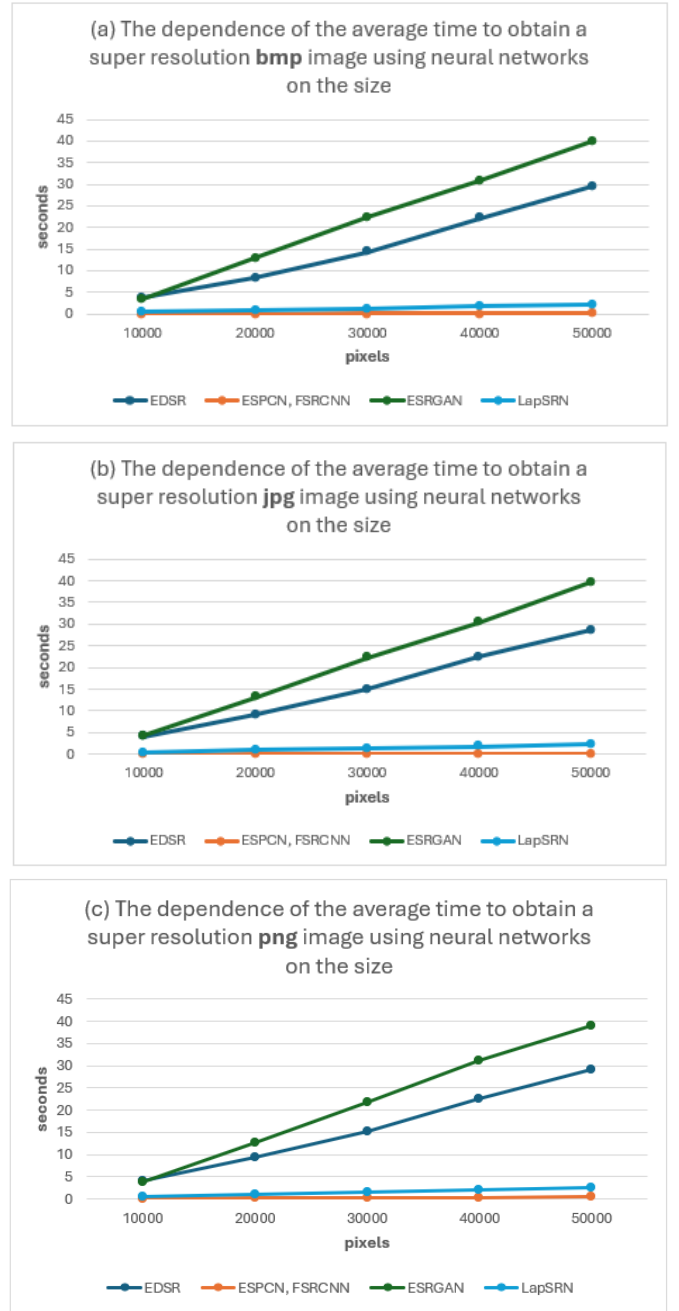


Fig. 2. The dependence of the average time to obtain a super resolution image using neural networks on the size. (a) bmp-format; (b) jpg-format; (c) png-format.

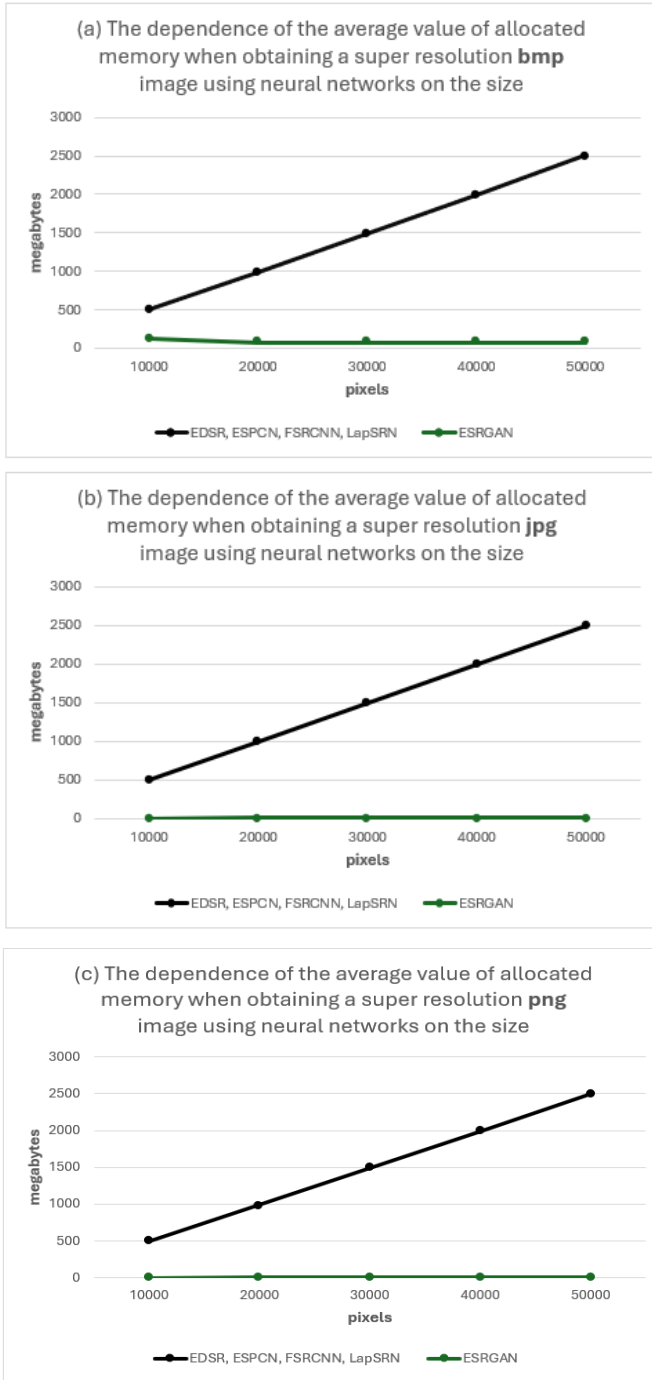


Fig. 3. The dependence of the average value of allocated memory when obtaining a super resolution image using neural networks on the size. (a) bmp-format; (b) jpg-format; (c) png-format.

As can be seen from the graphics, all analyzed neural networks for the selected formats have a linear $O(n)$ dependence [23] of the average time to obtain an improved image version on the photo size. ESPCN and FSRCNN show the best results.

The neural networks EDSR, ESPCN, FSRCNN, LapSRN also have a linear $O(n)$ dependence of the average allocated memory on the image size of all selected formats. ESRGAN has a constant dependence of $O(1)$ of the allocated memory for

jpg and png formats, and almost constant for bmp. The difference in memory allocation for ESRGAN when processing images in bmp format and 10000 size is since, presumably, this neural network is focused more on working with jpg and png, and additional resources are spent on primary photo processing. Also, significantly more memory is allocated for the bmp format (for ESRGAN) compared to other formats for the same reason. Nevertheless, ESRGAN shows significantly better results compared to other neural networks. The main reason is since the processing of the main part of the memory falls on the server located at the link [13].

The time/space efficiency is calculated by summing the ratio of the current time/space indicator to the minimum indicator among all neural networks for the current photo size and format, followed by dividing the product of the sizes (5) and formats (3) counts (total number of terms). Formulas 3 and 4 are presented below.

$$E_{T,i} = (\sum_{k=1}^F \sum_{j=1}^P t_{i,j,k} \text{avr} / t_{j,k} \text{avr min}) / (P * F) \quad (3)$$

where $E_{T,i}$ is the time efficiency of the neural networks i , $t_{i,j,k} \text{avr}$ is the average processing time of an image of size j and format k using the neural network i , $t_{j,k} \text{avr min}$ is the minimum average processing time of a photo of size j and format k among all neural networks, P is the sizes count (equal to 5), F is the formats count (equal to 3).

$$E_{S,i} = (\sum_{k=1}^F \sum_{j=1}^P s_{i,j,k} \text{avr} / s_{j,k} \text{avr min}) / (P * F) \quad (4)$$

where $E_{S,i}$ is the space efficiency of neural network i , $s_{i,j,k} \text{avr}$ is the average memory used when processing images of size j and format k using neural network i , $s_{j,k} \text{avr min}$ is the minimum average memory used when processing photos of size j and format k among all neural networks.

Let's assume that time and space indicators have the same impact. Then each component has the same weight, since there are 2 of them, the value is 0.5 ($=1/2$). Then the integral efficiency has the following formula (5):

$$E_{Int,i} = (E_{T,i} + E_{S,i} / 2) \quad (5)$$

where $E_{Int,i}$ is the integral efficiency of neural network i .

All efficiencies (time, space and integral) of the selected neural networks are presented in Table III. The values are rounded to 4 decimal places for time and space efficiencies, and to 5 digits for integral efficiencies.

TABLE III. THE EFFICIENCIES (TIME, SPACE AND INTEGRAL) OF THE NEURAL NETWORKS FOR IMAGE RESOLUTION IMPROVEMENT

Efficiencies	Neural network				
	EDSR	ESPCN	ESRGAN	FSRCNN	LapSRN
Time	0.0081	0.9558	0.0065	0.7964	0.0824
Space	0.0338	0.0338	1.0000	0.0338	0.0338
Integral	0.02095	0.49480	0.50325	0.4151	0.0581

As can be seen in the Table III, ESPCN has the best time efficiency (0.9558). ESRGAN is the best in space efficiency (1.0000). ESRGAN have the best integral efficiency (0.50325), ESPCN has a very close value (0.49480).

For a clearer representation of the results obtained in Table III, the efficiencies diagrams for all neural networks and their average values are plotted. Diagrams are presented in Fig. 4.

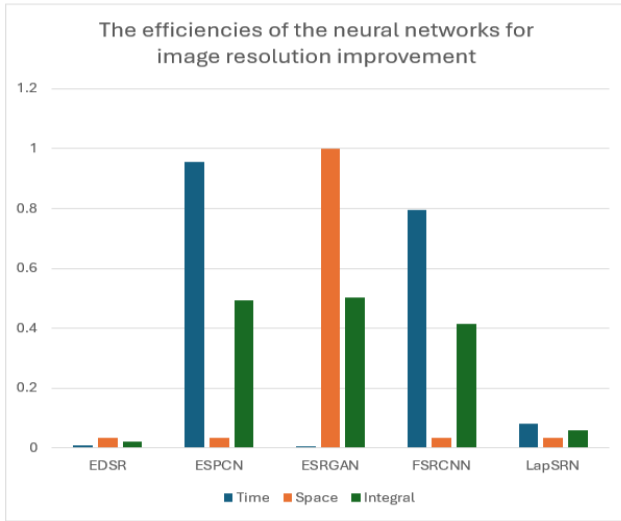


Fig. 4. The efficiencies diagrams of the neural networks for image resolution improvement.

In [6], the time and quality characteristics of the neural networks ESRGAN, FSRCNN and SRCNN are considered, but ESDR, ESPCN and LapSRN and the space component are not analyzed. Our work does not consider the SRCNN neural network, as well as quality indicator (image quality assessment is a relatively subjective process). The integral method described in the paper can be changed to a weight method, where weights are set by a user (instead of 0.5 for each of the characteristics, for example, the time weight is 0.6, and the space weight is 0.4).

V. CONCLUSION

In this paper, a comparative analysis of the neural networks efficiencies to improve image resolution, depending on size and format, using the integral approach is carried out. The neural networks selected are EDSR, ESPCN, ESRGAN, FSRCNN, LapSRN. The research material consists of road surface images in 5 sizes (100x100, 141x141, 173x173, 200x200, 224x224) and 3 formats (bmp, jpg, png). Each of the listed neural networks has a linear $O(n)$ dependence of the average time to obtain an improved image version on the photo size for all formats. EDSR, ESPCN, FSRCNN, LapSRN also have a linear $O(n)$ dependence of the average allocated memory on the size for all formats. ESRGAN has a constant $O(1)$ dependence on the allocated memory for jpg and png formats, and almost constant dependence for bmp. ESPCN has the best time efficiency (0.9558), and ESRGAN has the best space efficiency (1.0000). ESRGAN (0.50325) and ESPCN (0.49480) also have the best integral efficiency values.

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